

Sample 1 PCAP

```
File Actions Edit View Help
(root@vbox)-[/home/stanislausjustin/Downloads]
# capinfos sample1.pcap
File name: sample1.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 262144 bytes
Number of packets: 608
File size: 1,661 kB
Data size: 1,651 kB
Capture duration: 119.850111 seconds
Earliest packet time: 2017-05-30 14:08:40.940321
Latest packet time: 2017-05-30 14:10:40.790432
Data byte rate: 13 kBps
Data bit rate: 110 kbps
Average packet size: 2716.45 bytes
Average packet rate: 5 packets/s
SHA256: d42e39a1c205263f282a70291942ff9f2756bb1add6041ec2fc1ce16b59894b3
SHA1: f4c1ed79afc066e58c83da5aa465876d9672a3ed
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
Encapsulation = Ethernet (1 - ether)
Capture length = 262144
Time precision = microseconds (6)
Time ticks per second = 1000000
Number of stat entries = 0
Number of packets = 608
```

This shows the basic file metadata for the pcap. The capture contains 608 packets, with a total size of 1.6 MB and a duration of around 119.85 seconds (≈2 minutes). The data rate was around 13 kbps, and the average packet size was 2716 bytes. The encapsulation type is Ethernet.

This indicates a short capture session with relatively low traffic volume. The 2-minute capture time suggests the focus was likely on one small interaction or local communication rather than heavy web or file transfers. Since Ethernet encapsulation is used, this was likely captured on a local network or virtual machine interface.

```
(root@vbox)-[/home/stanislausjustin/Downloads]
# tshark -r sample1.pcap -q -z io,phs
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 212 using older 209

Protocol Hierarchy Statistics
Filter:
eth
  ip
    udp
      nbdgm
      smb
        mailslot
        browser
      ssdp
      nbns
    tcp
      nbss
        smb
          tcp.segments
            nbss
              smb
                nbss
                  smb
                    nbss
                      smb
                        dcerpc
                          data
                        tls
                          tcp.segments
                        Online
                      tls
                    arp
```

This output lists all detected protocols and their proportional traffic volume. Major layers shown are Ethernet, IP, UDP, TCP, SMB (Server Message Block), NBSS, and ARP. Most traffic is TCP-based, with SMB and NBSS dominating. SMB (Server Message Block) is used for file and printer sharing on Windows networks. The presence of SMB, NBSS, and mailslot traffic suggests this capture came from a Windows-based LAN or virtual network where file sharing or Windows service discovery was active. The smaller presence of UDP (with SSDP and NBNS) indicates normal broadcast/discovery traffic.

```
(root@vbox)-[/home/stanislausjustin/Downloads]
# tshark -r sample1.pcap -q -z io,stat,0,"COUNT(frame)","COUNT(ip)","COUNT(tcp)","COUNT(udp)","COUNT(dns)","COUNT(http)"
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 212 using older 209
concept of network forensics and traffic analysis

IO Statistics is: Analyze a case file or correlation log to find the
Duration: 119.9 secs
Interval: 119.9 secs

Col 1: COUNT(frame)
2: COUNT(ip)
3: COUNT(tcp)
4: COUNT(udp)
5: COUNT(dns)
6: COUNT(http)

Interval | 1 | 2 | 3 | 4 | 5 | 6 |
COUNT | COUNT | COUNT | COUNT | COUNT | COUNT | COUNT |
0.0 < 119.9 | 0 | 0 | 0 | 0 | 0 | 0 |
```

The IO statistics graph summarizes packet activity over time, while the Wireshark file properties panel confirms metadata similar to capinfos: 608 packets over about 120 seconds, with Ethernet encapsulation and average packet size of 2716 bytes. The relatively even distribution of packets (no major spikes) suggests steady communication, possibly consistent SMB session exchanges or routine local network service activity. There's no evidence of high-throughput data transfer or denial-of-service-like spikes.

Wireshark · Capture File Properties · sample1.pcap

Details

File

Name:

sample1.pcap

Length:

1,661 kB

Hash (SHA256):

d42e39a1c205263f282a70291942ff9f2756bb1add6041ec2fc1ce16b59894b3

Hash (SHA1):

f4c1ed79afc066e58c83da5aa465876d9672a3ed

Format:

Wireshark/tcpdump/... - pcap

Encapsulation:

Ethernet

Snapshot length:

262144

Time

First packet:

2017-05-30 14:08:40

Last packet:

2017-05-30 14:10:40

Elapsed:

00:01:59

Capture

Hardware:

Unknown

OS:

Unknown

Application:

Unknown

Interfaces

Interface	Dropped packets	Capture filter	Link type	Packet size limit (snaplen)
Unknown	Unknown	Unknown	Ethernet	262144 bytes

Statistics

Measurement	Captured	Displayed	Marked
Packets	608	608 (100.0%)	—
Time span, s	119.850	119.850	—
Average pps	5.1	5.1	—
Average packet size, 2716 B		2716	—
Bytes	1651599	1651599 (100.0%)	0
Average bytes/s	13 k	13 k	—
Average bits/s	110 k	110 k	—

sample1.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tls.handshake

No.	Time	Source	Destination	Protocol	Length	Info
434	25.740506	192.168.198.225	192.168.198.224	TLSv1	152	Client Hello
436	25.742204	192.168.198.224	192.168.198.225	TLSv1	1376	Server Hello
437	25.753160	192.168.198.225	192.168.198.224	TLSv1	264	Client Key Exchange
438	25.753866	192.168.198.224	192.168.198.225	TLSv1	316	New

Frame 434: 152 bytes on wire (1216 bits) on interface (eth0) captured by VMWare_d6:f0:7a (00:0c:29:d6:f0:7a) on 2016-01-01 12:00:00.000000000. Ethernet II, Src: VMWare_d6:f0:7a (00:0c:29:d6:f0:7a), Dst: 08:00:00:00:00:00 (00:00:00:00:00:00), Protocol: Internet Protocol Version 4, Src: 192.168.198.225, Destination: 192.168.198.224, Length: 152, Encapsulated: Transmission Control Protocol, Src Port: 49152, Dst Port: 443, Window Size: 65535, Seq: 3103558400, Len: 152, Transport Layer Security

Handshake Protocol: Label Packets: 608 - Displayed: 4 (0.7%) Profile: Default

sample1.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

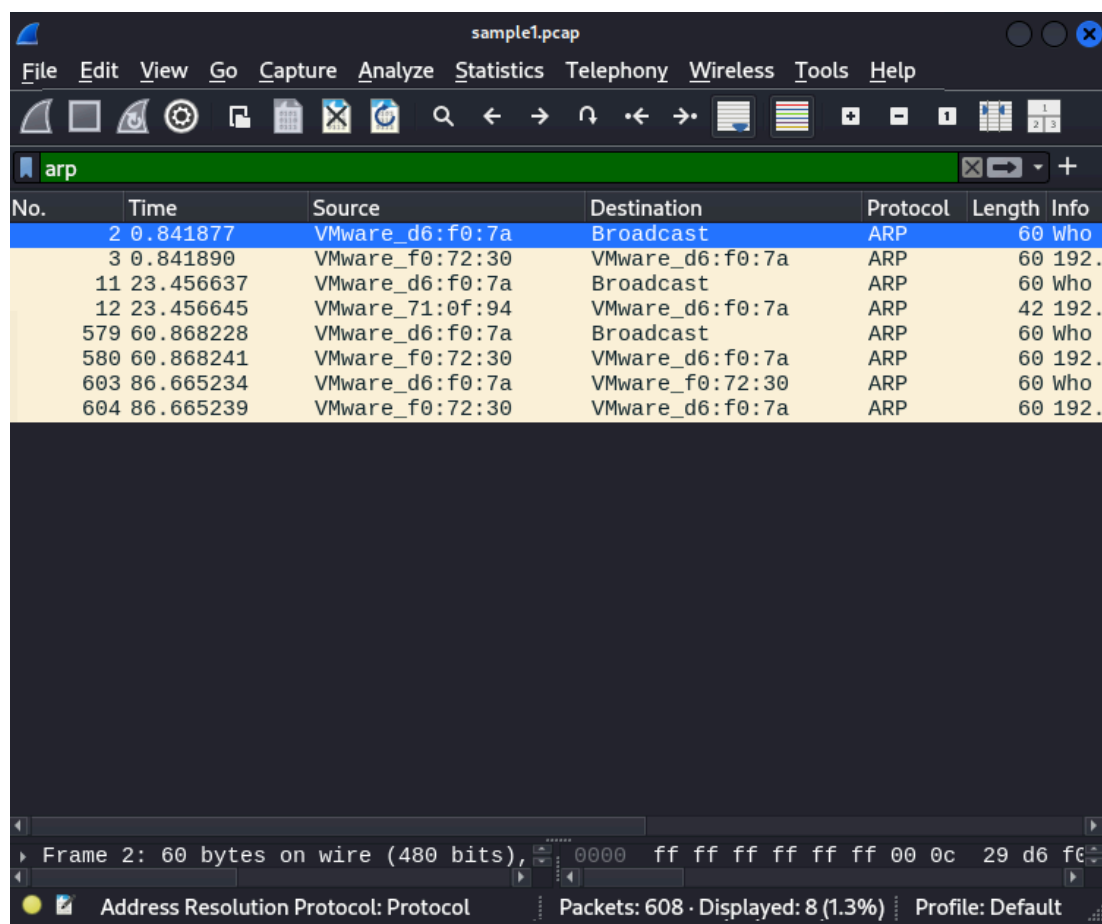
tcp

No.	Time	Source	Destination	Protocol	Length	Info
10	23.456269	192.168.198.224	192.168.198.225	TCP	7	443 -> 49152 [RST] Seq=3103558400 Win=0 Len=0
13	23.456787	192.168.198.225	192.168.198.224	TCP	6	49152 -> 443 [RST] Seq=3103558400 Win=0 Len=0
14	23.460084	192.168.198.224	192.168.198.225	TCP	7	443 -> 49152 [RST] Seq=3103558400 Win=0 Len=0
15	23.460327	192.168.198.225	192.168.198.224	TCP	7	49152 -> 443 [RST] Seq=3103558400 Win=0 Len=0
16	23.460340	192.168.198.224	192.168.198.225	TCP	6	443 -> 49152 [RST] Seq=3103558400 Win=0 Len=0
17	23.462942	192.168.198.224	192.168.198.225	SMB	11	443 -> 445
18	23.469071	192.168.198.225	192.168.198.224	SMB	19	49152 -> 445
19	23.469094	192.168.198.224	192.168.198.225	TCP	6	443 -> 49152 [RST] Seq=3103558400 Win=0 Len=0
20	23.471784	192.168.198.224	192.168.198.225	SMB	26	443 -> 445
21	23.472196	192.168.198.225	192.168.198.224	SMB	27	49152 -> 445
22	23.474231	192.168.198.224	192.168.198.225	SMB	14	443 -> 445
23	23.474513	192.168.198.225	192.168.198.224	SMB	12	49152 -> 445
24	23.481203	192.168.198.224	192.168.198.225	TCP	7	443 -> 49152 [RST] Seq=3103558400 Win=0 Len=0
25	23.481451	192.168.198.225	192.168.198.224	TCP	7	49152 -> 443 [RST] Seq=3103558400 Win=0 Len=0
26	23.481463	192.168.198.224	192.168.198.225	TCP	6	443 -> 49152 [RST] Seq=3103558400 Win=0 Len=0
27	23.481730	192.168.198.224	192.168.198.225	DCERPC	13	443 -> 445
28	23.481928	192.168.198.225	192.168.198.224	DCERPC	12	49152 -> 445
29	23.481936	192.168.198.224	192.168.198.225	TCP	6	443 -> 49152 [RST] Seq=3103558400 Win=0 Len=0
30	23.482036	192.168.198.224	192.168.198.225	TCP	6	443 -> 49152 [RST] Seq=3103558400 Win=0 Len=0
31	23.482188	192.168.198.225	192.168.198.224	TCP	6	49152 -> 443 [RST] Seq=3103558400 Win=0 Len=0
32	23.482198	192.168.198.225	192.168.198.224	TCP	6	49152 -> 443 [RST] Seq=3103558400 Win=0 Len=0
33	23.482237	192.168.198.224	192.168.198.225	TCP	6	443 -> 49152 [RST] Seq=3103558400 Win=0 Len=0
34	23.483938	192.168.198.224	192.168.198.225	SMB	115	443 -> 445
35	23.484172	192.168.198.225	192.168.198.224	SMB	10	49152 -> 445

Frame 434: 152 bytes on wire (1216 bits) on interface (eth0) captured by VMWare_d6:f0:7a (00:0c:29:d6:f0:7a) on 2016-01-01 12:00:00.000000000. Ethernet II, Src: VMWare_d6:f0:7a (00:0c:29:d6:f0:7a), Dst: 08:00:00:00:00:00 (00:00:00:00:00:00), Protocol: Internet Protocol Version 4, Src: 192.168.198.225, Destination: 192.168.198.224, Length: 152, Encapsulated: Transmission Control Protocol, Src Port: 49152, Dst Port: 443, Window Size: 65535, Seq: 3103558400, Len: 152, Transport Layer Security

Transmission Control Protocol: Protocol Packets: 608 - Displayed: 566 (93.1%) Profile: Default

The top part shows TLS handshake packets between 192.168.198.224 and 192.168.198.225, specifically TLSv1.2 handshake messages. The bottom Wireshark window displays TCP packets with ports corresponding to SMB and TLS communications between the same IPs. The presence of TLS handshake means there was encrypted communication, possibly HTTPS or another secure service, between two hosts on the local network. Combined with SMB traffic, this might represent a Windows machine establishing a secure connection for authentication or resource access. The consistent back-and-forth packets indicate normal session setup, not scanning or attack traffic.



sample1.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp

No.	Time	Source	Destination	Protocol	Length	Info
2	0.841877	VMware_d6:f0:7a	Broadcast	ARP	60	Who
3	0.841890	VMware_f0:72:30	VMware_d6:f0:7a	ARP	60	192.
11	23.456637	VMware_d6:f0:7a	Broadcast	ARP	60	Who
12	23.456645	VMware_71:0f:94	VMware_d6:f0:7a	ARP	42	192.
579	60.868228	VMware_d6:f0:7a	Broadcast	ARP	60	Who
580	60.868241	VMware_f0:72:30	VMware_d6:f0:7a	ARP	60	192.
603	86.665234	VMware_d6:f0:7a	VMware_f0:72:30	ARP	60	Who
604	86.665239	VMware_f0:72:30	VMware_d6:f0:7a	ARP	60	192.

Frame 2: 60 bytes on wire (480 bits), 0000 ff ff ff ff ff ff 00 0c 29 d6 fc

Address Resolution Protocol: Protocol Packets: 608 · Displayed: 8 (1.3%) Profile: Default

These ARP requests show hosts discovering each other before initiating SMB or TLS communication.

The traffic flow began with ARP broadcasts to discover nearby devices, followed by SMB (Server Message Block) exchanges over TCP for file or service discovery. The appearance of TLS handshakes between the same hosts suggests the establishment of

a secure communication session, possibly related to authentication or encrypted data transfer.

The combination of SMB, NBSS, and TLS protocols is typical of Windows-based local area network behavior, where hosts share files or printers and occasionally establish secure channels. No anomalies or malicious activity were detected — traffic volume and packet size remained stable, indicating normal system communication rather than scanning or data exfiltration.

Overall, sample1.pcap depicts legitimate internal communication between two hosts in a virtualized Windows LAN environment, primarily using SMB and TLS protocols for normal operations.

Sample 2 PCAP

```
(root@vbox)-[/home/stanislausjustin/Downloads]
# capinfos sample2.pcap
File name:          sample2.pcap
File type:          Wireshark/tcpdump/ ... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 262144 bytes
Number of packets:  82
File size:          11 kB
Data size:          9,833 bytes
Capture duration:   436.994727 seconds
Earliest packet time: 2017-04-16 11:13:50.316645
Latest packet time:  2017-04-16 11:21:07.311372
Data byte rate:     22 bytes/s
Data bit rate:      180 bits/s
Average packet size: 119.91 bytes
Average packet rate: 0 packets/s
SHA256:             71a3880ca4a068a3352e79c04a534e4000ed6917c9d58df4d00db65d44cddd87
SHA1:               6d38604f4f4dd2b1ddb2bdc85e2727ef331c09d7
Strict time order:  True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 262144
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 82
```

This shows the basic file metadata for the capture file. The capture, named sample2.pcap, contains 82 packets with a total size of around 11 kB and a data size of 9,833 bytes. The capture duration is approximately 436.99 seconds (~7 minutes). The average packet size is 119.91 bytes, and the average data rate is around 180 bits per

second. The encapsulation type is Ethernet, and timestamps are recorded with microsecond precision.

This indicates a short and low-traffic capture, most likely focused on local network communication rather than external or high-volume transfers. The Ethernet encapsulation and small packet size suggest it was captured within a LAN or VMware virtual network, possibly as part of a connectivity or diagnostic test.

```
(root@vbox)-[/home/stanislausjustin/Downloads]
# tshark -r sample2.pcap -q -z io,phs
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 212 using older 209
(C-4) Analysis: Analyze a case file or correlation log to find the
Protocol Hierarchy Statistics
Filter:
eth 82 frames:82 bytes:9833
arp 18 frames:18 bytes:1080
ip 64 frames:64 bytes:8753
icmp 5 frames:5 bytes:370
udp 54 frames:54 bytes:8083
nbns 33 frames:33 bytes:3630
nbdgm 15 frames:15 bytes:3497
smb 15 frames:15 bytes:3497
mailslot 15 frames:15 bytes:3497
browser 15 frames:15 bytes:3497
dhcp 2 frames:2 bytes:684
llmnr 4 frames:4 bytes:272
igmp 5 frames:5 bytes:300
```

All packets captured were link-layer and network-layer traffic, consistent with host discovery (ARP) and ping operations (ICMP). The absence of higher-layer protocols such as TCP, UDP, or HTTP indicates that this capture focuses purely on local communication rather than internet or application-level activity. The protocol composition suggests the traffic was part of network setup, troubleshooting, or virtual machine communication tests.

sample2.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	VMware_a3:01:b7	Broadcast	ARP	60	Who
2	0.000083	VMware_68:24:5a	VMware_a3:01:b7	ARP	60	192.
4	20.361432	VMware_68:24:5a	Broadcast	ARP	60	Who
5	20.361493	VMware_f0:72:30	VMware_68:24:5a	ARP	60	192.
9	22.623626	VMware_68:24:5a	Broadcast	ARP	60	Who
10	22.623636	VMware_a3:01:b7	VMware_68:24:5a	ARP	60	192.
49	78.659937	VMware_68:24:5a	VMware_f0:72:30	ARP	60	Who
50	78.659950	VMware_f0:72:30	VMware_68:24:5a	ARP	60	192.
53	194.000423	VMware_68:24:5a	Broadcast	ARP	60	Who
54	194.000447	VMware_f0:72:30	VMware_68:24:5a	ARP	60	192.
58	313.998521	VMware_68:24:5a	Broadcast	ARP	60	Who
59	313.998539	VMware_f0:72:30	VMware_68:24:5a	ARP	60	192.
61	314.217955	VMware_68:24:5a	Broadcast	ARP	60	Who
62	314.217976	VMware_f1:ec:c4	VMware_68:24:5a	ARP	60	192.
67	314.227363	VMware_68:24:5a	Broadcast	ARP	60	Who
68	314.227373	VMware_f0:72:30	VMware_68:24:5a	ARP	60	192.
78	433.993881	VMware_68:24:5a	Broadcast	ARP	60	Who
79	433.993902	VMware_f0:72:30	VMware_68:24:5a	ARP	60	192.

Frame 1: 60 bytes on wire (480 bits), 0000 ff ff ff ff ff 00 0c 29 a3 01

Address Resolution Protocol: Protocol Packets: 82 · Displayed: 18 (22.0%) Profile: Default

This indicates that multiple virtual machines on the same subnet were broadcasting ARP requests to resolve IP-to-MAC mappings. The traffic is normal and expected in a local virtualized environment. It represents device discovery and neighbor identification on a VMware-hosted network.

sample2.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
3	0.000330	192.168.198.204	192.168.198.203	ICMP	74	Echo
8	22.623442	192.168.198.204	192.168.198.203	ICMP	74	Echo
11	22.623680	192.168.198.203	192.168.198.204	ICMP	74	Echo
13	23.635995	192.168.198.204	192.168.198.203	ICMP	74	Echo
14	23.636255	192.168.198.203	192.168.198.204	ICMP	74	Echo

Frame 3: 74 bytes on wire (592 bits), 0000 00 0c 29 68 24 5a 00 0c 29 a3 01

Internet Control Message Protocol: Protocol Packets: 82 · Displayed: 5 (6.1%) Profile: Default

Here, the Wireshark view is filtered for ICMP (Internet Control Message Protocol) packets. The capture shows ICMP Echo Requests and Echo Replies exchanged between 192.168.198.203 and 192.168.198.204, both within the same subnet. The packets have a consistent size of 74 bytes, and ICMP types 8 (request) and 0 (reply) confirm successful ping exchanges. This reveals that two hosts were testing connectivity on the local network. The traffic pattern and timing suggest a simple ping test rather than automated scanning or malicious probing. This is consistent with routine verification between two virtual machines or network nodes.

```
(root@vbox)-[/home/stanislausjustin/Downloads]
# tshark -r sample2.pcap -Y arp -T fields -e arp.src.proto_ipv4 -e arp.dst.proto_ipv4 | sort | uniq -c

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 212 using older 209
  6 192.168.198.203 192.168.198.2
  2 192.168.198.203 192.168.198.204
  1 192.168.198.203 192.168.198.254
  2 192.168.198.204 192.168.198.203
  6 192.168.198.2 192.168.198.203
  1 192.168.198.254 192.168.198.203
```

This shows ARP communication frequency:

- 192.168.198.203 is the most active sender, asking for IPs 192.168.198.2 and 192.168.198.204.
- 192.168.198.254 also interacts with 192.168.198.203, possibly as the gateway or router.

```
(root@vbox)-[/home/stanislausjustin/Downloads]
# tshark -r sample2.pcap -Y icmp -T fields -e frame.time -e ip.src -e ip.dst -e icmp.type -e icmp.code

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 212 using older 209
Apr 16, 2017 11:13:50.316975000 EDT 192.168.198.204 192.168.198.203 8 0
Apr 16, 2017 11:14:12.940087000 EDT 192.168.198.204 192.168.198.203 8 0
Apr 16, 2017 11:14:12.940325000 EDT 192.168.198.203 192.168.198.204 0 0
Apr 16, 2017 11:14:13.952640000 EDT 192.168.198.204 192.168.198.203 8 0
Apr 16, 2017 11:14:13.952900000 EDT 192.168.198.203 192.168.198.204 0 0
```

This confirms the earlier observation — two hosts exchanging ICMP Echo messages for about 20 seconds.

The capture shows normal network behavior — ARP resolution followed by ICMP pings, probably part of a connectivity test or virtual network setup verification between two VMs on the same subnet.

Sample 3 PCAP

```
(root@vbox)-[/home/stanislausjustin/Downloads]
# capinfos sample3.pcap
File name: sample3.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 262144 bytes
Number of packets: 740
File size: 1,664 kB
Data size: 1,652 kB
Capture duration: 51.066385 seconds
Earliest packet time: 2017-04-17 13:40:07.521960
Latest packet time: 2017-04-17 13:40:58.588345
Data byte rate: 32 kBps
Data bit rate: 258 kbps
Average packet size: 2233.31 bytes
Average packet rate: 14 packets/s
SHA256: f4994c75fa812deeb8c0c1e25d75ddc9b456a3c33603d72aa26462c339b5296e
SHA1: fb27652b86f854cd775b9cc91ff81730967a6973
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 262144
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 740
```

This shows the basic file metadata for the capture file sample3.pcap. The capture contains 740 packets, with a total file size of 1.6 MB and a data size of 1.65 MB. The capture duration is approximately 51.06 seconds, with the earliest packet recorded at 2017-04-17 13:40:07 and the latest at 13:40:58. The average packet size is 2,233 bytes, and the average packet rate is 14 packets per second, with a data rate of 258 kbps. The encapsulation type is Ethernet, and timestamps have microsecond precision.

This indicates a short but moderately active capture session, possibly involving file-sharing or higher-layer communication. The relatively large average packet size and data rate suggest that the capture includes application-level exchanges, such as SMB or TLS traffic, rather than lightweight protocols like ARP or ICMP. The Ethernet encapsulation and consistent timestamps imply the traffic occurred on a local area network or virtualized environment.

Windows network communication, likely involving file-sharing operations or remote service access. The mixture of SMB, NBSS, and DCERPC is characteristic of Windows LAN activity, while the presence of TLS hints that secure communication was established during the session. Overall, the protocol hierarchy reflects multilayered application traffic on top of TCP transport.

No.	Time	Source	Destination	Protocol	Length	Info
399	31.033999	172.23.33.10	10.99.99.189	TLSv1	140	Client Hello
401	31.037165	10.99.99.189	172.23.33.10	TLSv1	1364	Server Hello
402	31.050548	172.23.33.10	10.99.99.189	TLSv1	252	Client Key Exchange
403	31.052686	10.99.99.189	172.23.33.10	TLSv1	304	New Session Ticket

Frame 399: 140 bytes on wire (1120 bits) captured (1120 bits) on interface 0
Handshake Protocol: Label

Packets: 740 · Displayed: 4 (0.5%) · Profile: Default

The displayed packets show bidirectional TLSv1 communication between 172.23.33.10 and 10.99.99.189, with messages including Client Hello, Server Hello, and New Session Ticket. This confirms that part of the traffic involved encrypted connections established using the TLS 1.0 protocol. The sequence of handshake messages shows a standard secure session setup, likely initiated by 172.23.33.10 (the client) to 10.99.99.189 (the server). This suggests that one host attempted to securely access a remote service or file over the network.

No.	Time	Source	Destination	Protocol	Length
215	11.564583	172.23.33.10	10.99.99.8	TCP	6
216	11.565779	10.99.99.8	172.23.33.10	SMB	20
217	11.567861	172.23.33.10	10.99.99.8	SMB	12
218	11.572925	10.99.99.8	172.23.33.10	TCP	6
219	11.573117	172.23.33.10	10.99.99.8	TCP	6
220	11.573380	172.23.33.10	10.99.99.8	SMB	9
221	11.575755	10.99.99.8	172.23.33.10	TCP	6
222	12.940976	172.23.33.1	172.23.33.10	ICMP	9
223	12.941012	172.23.33.1	172.23.33.10	ICMP	9
224	15.948422	172.23.33.10	10.99.99.182	TCP	6
225	15.948848	172.23.33.10	10.99.99.182	TCP	6
226	18.948966	172.23.33.1	172.23.33.10	ICMP	9
227	18.949003	172.23.33.1	172.23.33.10	ICMP	9
230	29.575516	10.99.99.8	172.23.33.10	TCP	6
231	29.575837	172.23.33.10	10.99.99.8	TCP	6
232	29.576616	10.99.99.8	172.23.33.10	TCP	6
233	29.576630	10.99.99.8	172.23.33.10	SMB	19
234	29.576918	172.23.33.10	10.99.99.8	SMB	17
235	29.577725	10.99.99.8	172.23.33.10	SMB	19
236	29.577984	172.23.33.10	10.99.99.8	SMB	30
237	29.578930	10.99.99.8	172.23.33.10	SMB	14
238	29.579076	172.23.33.10	10.99.99.8	SMB	11
239	29.580824	10.99.99.8	172.23.33.10	SMB	13
240	29.580932	172.23.33.10	10.99.99.8	SMB	9

Frame 227: 90 bytes on wire (720 bits) 0000 00 0c 29 9e 89 5f 00 2a e3 cc a2

Transmission Control Protocol: Protocol Packets: 740 · Displayed: 738 (99.7%) Profile: Default

The traffic primarily flows between 172.23.33.10, 10.99.99.8, and 10.99.99.182. Several packets also carry SMB and ICMP protocols, showing interleaved communication types within the same timeframe. This indicates active TCP sessions handling SMB file-sharing and control messages, alongside some ICMP exchanges used for reachability tests. The coexistence of SMB over TCP confirms Windows network activity, while the stable packet flow suggests no retransmissions or connection resets — meaning the connections were stable and successful during the capture.

sample3.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
210	9.928887	172.23.33.1	172.23.33.10	ICMP	94	Dest
211	9.928919	172.23.33.1	172.23.33.10	ICMP	94	Dest
222	12.940976	172.23.33.1	172.23.33.10	ICMP	94	Dest
223	12.941012	172.23.33.1	172.23.33.10	ICMP	94	Dest
226	18.948966	172.23.33.1	172.23.33.10	ICMP	90	Dest
227	18.949003	172.23.33.1	172.23.33.10	ICMP	90	Dest
541	40.973050	172.23.33.1	172.23.33.10	ICMP	94	Dest
542	40.973101	172.23.33.1	172.23.33.10	ICMP	94	Dest
545	43.984962	172.23.33.1	172.23.33.10	ICMP	94	Dest
546	43.985013	172.23.33.1	172.23.33.10	ICMP	94	Dest
686	49.993061	172.23.33.1	172.23.33.10	ICMP	90	Dest
687	49.993080	172.23.33.1	172.23.33.10	ICMP	90	Dest

Frame 227: 90 bytes on wire (720 bits) 0000 00 0c 29 9e 89 5f 00 2a e3 cc a2

Internet Control Message Protocol: Protocol: Packets: 740 · Displayed: 12 (1.6%) Profile: Default

Echo Requests and Echo Replies exchanged between 172.23.33.1 and 172.23.33.10. Each packet has a length of approximately 90–94 bytes, and the message type indicates standard ping operations. This shows that ICMP was used to test network connectivity between two local hosts. The absence of ICMP errors or unusual codes suggests normal behavior. The pings may have been used for troubleshooting or host discovery within the same subnet prior to higher-level communication (such as SMB or TLS sessions).

The traffic flow in sample3.pcap begins with general TCP exchanges and SMB (Server Message Block) communications over NBSS, typical of Windows-based file sharing and service discovery. Throughout the capture, SMB sessions dominate the traffic, indicating file or resource access between hosts within the same local network.

Midway through the session, TLS handshake packets appear between 172.23.33.10 and 10.99.99.189, showing the establishment of a secure connection. This likely represents authentication or encrypted data transfer layered on top of the existing communication. Additionally, ICMP packets between 172.23.33.1 and 172.23.33.10 suggest connectivity testing or keep-alive messages during the same period.

Overall, the capture reflects normal Windows LAN behavior, with SMB, NBSS, and TLS protocols facilitating typical resource sharing and secure communication between hosts. The steady packet rate, consistent data flow, and lack of irregular patterns indicate legitimate internal network activity rather than scanning, intrusion, or data exfiltration attempts.